

Order Flow Imbalances

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Question 1: What's the motivation behind measuring OFI at multiple depth levels of the order book?

Firstly, the best-level OFI, while informative, offers only a partial view by capturing the most immediate liquidity. Deeper levels contain valuable information about latent liquidity provision and demand, reflecting intentions that may influence future price movements. The study demonstrates that incorporating multi-level information significantly enhances explanatory power for price dynamics; the authors construct an integrated OFI using PCA which increases the in-sample R^2 from approximately 71% for best-level OFI to 87%. Furthermore, the relative importance of different levels can vary based on stock characteristics like volume and volatility, reinforcing the need to look beyond the top level.

Secondly, different market participants employ strategies involving various depths. Market makers might place passive orders deeper in the book to manage inventory, while informed traders could use multi-level strategies to execute large orders with minimal impact. Aggregating OFI across levels captures the net effect of these diverse strategies.

Thirdly, the best bid and ask are often the most volatile levels, susceptible to noise from fleeting orders or potential manipulation tactics (e.g., quote flickering). Deeper levels tend to exhibit more stability, potentially reflecting more committed trading interest and providing a clearer signal of underlying supply-demand pressure.

Finally, orders naturally transition between levels as prices fluctuate and the book evolves. Measuring OFI across multiple depths captures this dynamic interplay and provides richer insight into the price formation process than a single-level view allows.

Question 2: Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The cross-impact model potentially involves a large number of predictors (e.g. the OFIs of many other assets). Estimating such a model with OLS carries a significant risk of overfitting, especially if the predictor variables (cross-asset OFIs) exhibit correlation. Lasso addresses this through L_1 regularization, which penalizes the sum of the absolute values of the coefficients. This penalty encourages sparsity by shrinking many coefficients exactly to zero.

Sparsity is desirable both theoretically and empirically in this context. Financial intuition suggests that only a subset of assets (e.g., those within the same sector or economically linked) should exert a meaningful cross-impact on a given stock. Lasso automatically performs variable selection, identifying potentially significant cross-impact relationships while discarding less relevant ones. The authors' empirical results confirm this, showing a sparse cross-impact matrix.

Furthermore, financial market data often suffers from multicollinearity, where predictors are correlated. OLS estimates can become unstable and unreliable under multicollinearity. Lasso is more robust in such situations, as it tends to select one variable from a group of highly correlated variables and shrink the others towards zero.

Additionally, in high-frequency settings, the number of potential predictors ($N \approx 100$) can easily exceed the number of observations within a short estimation window (e.g., 30 minutes of 1-minute data yields

30 observations). In such cases, OLS is mathematically ill-posed (more predictors than observations). Lasso, as a regularized method, can still provide a stable solution.

Finally, the resulting sparse model from Lasso is often more interpretable, highlighting the most influential cross-asset relationships identified by the data.

Question 3: Why is OFI considered a better predictor of short-term returns than trade volume?

Order Flow Imbalance (OFI) is generally considered a superior predictor of short-term returns compared to simple trade volume because it captures more nuanced and directional information about supply and demand dynamics within the limit order book.

The fundamental difference lies in directionality. Trade volume quantifies the level of activity but is agnostic to whether the activity represents buying or selling pressure. OFI, by construction, measures the net pressure, the imbalance between aggressive buying (lifting the ask or adding to the bid) and aggressive selling (hitting the bid or adding to the ask). This directional component is crucial for anticipating short-term price movements, as emphasized in foundational research stating that price changes over short intervals are mainly driven by OFI.

OFI incorporates information beyond just executed trades. Its calculation typically includes changes in limit order sizes and prices at various levels of the book. This captures shifts in unexecuted intentions reflected in the LOB (e.g. new orders being placed, existing orders being cancelled or modified), which often precede actual trades and contain valuable forward-looking information. Since limit order book events vastly outnumber actual trades, OFI leverages a much richer information set than volume alone.

OFI inherently reflects liquidity dynamics by accounting for the balance between liquidity consumption (market orders) and liquidity provision/withdrawal (limit order changes). Volume only reflects the consumptive part (trades).

Empirical evidence presented in the study shows that lagged OFI, particularly cross-asset OFI, contains predictive power for future returns over short horizons (minutes), acting as a leading indicator. While volume might correlate with volatility, its directional predictive power is generally weaker than OFI's.